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Prototypes as Instruments for Critical Practice

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In an era of accelerating technological development, climate urgency, and shifting paradigms of architectural research, the prototype emerges as a vital instrument – both as a medium of inquiry and as a mode of expression. This issue of Dimensions explores and questions the multifaceted nature of prototypes, emphasizing and interrogating their capacity to generate, embody, and communicate knowledge across scales, activities, and disciplinary boundaries.

The issue is configured as a sequence of research papers and interviews, all addressing, examining, and discussing a contemporary understanding and use of prototypes. The research papers document tests of materials, construction methods, and architectural frameworks; the interviews represent conversations and insights gained from academics and practitioners. The two formats acknowledge different modes of engagement in academia and in practice, pointing to the need to understand, develop, and in some cases, bridge the two.

The term »prototype« is interpreted and represented broadly to encompass early-stage developments, demonstrators, mock-ups, biopsies, and refined constructs. The prototypes examined in this issue encapsulate an entire design concept or isolate a specific aspect of inquiry. They are not merely technical artifacts but epistemic instruments – vehicles through which designers and researchers discover, test, reveal, and refine ideas.

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Prototyping is situated here within a broader framework of critical practice, understood as the avant-garde of method and knowledge creation. Intentionally, no existing framework, dichotomy, or taxonomy of prototypes has been formulated or requested by contributors. Instead, the issue seeks to openly offer perspectives and practices that add further understanding, nuance, and characteristics to the term, from which new paths can be started and combined.

The contributions operate within both basic and applied research contexts, including research by design and collaborative academic–industry initiatives. The research shared includes the examination of new materials or the use of historic structural frameworks. Lorenzo Renzullo examines the developments of Italian facades along the lines of prototypes and mock-ups in the middle of the 20th century, documenting the productive role of the full-scale physical mock-up. Deeply immersed in testing the capacity of old and new materials, Matthias Ballestrem and Summer Islam work with paludiculture-based materials, while Mahan Motalebi and Marcus Shaffer examine intentional vulnerability in earthen construction. Douglas Sadler, Thomas Udin, and Christina Godiksen construct Guastavino vaults from laminated marble slabs taken from quarry waste piles. Also working with the remnants of construction, in this case, waste wood from mass timber projects, Miriam Dunn tests different frameworks of circular construction for timber housing units. Dinorah Schulte and Edurne Morales merge traditional Mexican craft with digital fabrication to discover through prototyping that they can enrich each other. In many instances, the prototype becomes a site of intensive discourse – where considerations of economy, longevity, liability, and maintenance intersect with aesthetic, ethical, and environmental imperatives. Anke Pasold and Isak Worre Foged demonstrate that friction is a productive force in their essay through a persistent development and testing of the Paulownia House, formulated as a *Continuum – Poetics of Prototypes*. Pedram Ghelichi probes the process of prototyping a plywood system during an iterative, participatory construction process and the concurrent evolution of the *vve system*.

The series of interviews invites and questions the role and instrumentality of prototypes in architectural practices that have demonstrated a desire and commitment to innovate architectural thinking, design, and construction. These practices represent a range from internationally established and renowned offices to small studios with integrated prototyping

workshops and one-person pursuits of novelty through direct experimental making.

Pål Hasselberg, senior architect at Snøhetta, reveals how the office works with clients through specific collaborative processes to develop conceptual, instrumental, and practical prototypes. Similarly, Simeon Balford, partner at Reværk, articulates the in-situ findings and idea generation that can take place with a client through prototype activities, where omitting the physical investigation would result in a lack of advancement and a return to known solutions.

These conditions are also recognized in the practice of Dorte Mandrup Architects, where senior partner Kasper Pilemand reveals how prototyping activities are both stringed together and form an open ecology of tools that can be accessed to advance early ideas and build confidence in maintaining complex design proposals.

In the hands of Tom Svilans, the prototype hovers between understanding, describing, and activating the complexity of wood structures, where fabrication technology, production processes, and expressive characteristics seem to be integral to the prototype under question. Similarly, Kory Bieg oscillates between academic and industry modes of thinking and using the prototype, connecting his own practice and research to didactic ideas and teaching activities.

Across articles and interviews, it becomes clear that the conflicts identified are not seen as obstacles but as catalysts for conceptual and practical innovation. They provoke reflection and recalibration, enabling the emergence of new forms of architectural knowledge and transfer to building practices.

Prototypes function as instruments that advance collaborative methodologies and research strategies. They facilitate synergetic engagements across disciplinary and institutional boundaries, enabling situated analysis and prescriptive observation through making. Whether physical, digital, or hybrid, prototypes embody specific investigations and articulate distinct contributions to the field.

The urgency of environmental degradation and material exhaustion underscores the need for prototyping practices that critically engage with sustainability. Prototypes are increasingly tasked with addressing systemic challenges – rethinking material cycles, reusing existing resources, and restructuring design paradigms.

This editorial invites reflection on the epistemological dimensions of prototyping. Key questions included:

- What forms of knowledge are embodied in the prototype or emerge through its development?
- How do prototypes mediate collaboration between academic and professional domains?
- In what ways do prototypes refine or redefine applied research strategies?

Multiple answers are given by the contributions. By foregrounding the prototype as a critical instrument, this issue contributes to an expanded understanding of architectural research – one that privileges experimentation, reflection, and the generative potential of making.